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Not All Spines Are Created Equal: How CT Segmenters Fail on Lumbosacral Transitional Vertebrae

 (/pdf?id=D8yRWDAGQF)

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 Revisions (/revisions?id=D8yRWDAGQF)  BibTeX

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Keywords:  CT segmentation, lumbosacral transitional vertebra, LSTV, wrong-level surgery, spine surgical planning, nnU-Net, TotalSegmentator, transitional vertebra labeling, vertebra counting, spinopelvic anatomy, Castellvi classification, sacrum segmentation, hip segmentation, medical image segmentation, CTSpine1K, CTPelvic1K, COLONOG, partial-annotation supervision, LSTV-stratified evaluation, junction Dice, surgical safety, never-event prevention, deep learning for radiology, benchmark dataset

TL;DR: TotalSegmentator silently mislabels LSTV cases one level off—reproducing the wrong-level surgery error. We release CTSpinoPelvic1K (LSTV-stratified, Castellvi-typed) and the first openly-released dual spinopelvic CT segmenter.

Abstract:

Wrong-level spine surgery remains a persistent never-event, with the Lumbosacral Transitional Vertebra (LSTV, 5–35% prevalence) as its largest anatomic driver. We audit TotalSegmentator — the most widely deployed CT segmentation system and default back-end of 3D Slicer — and show that on lumbarization cases its output is anatomically indistinguishable from a wrong-level surgical plan: lacking an L6 class, it shifts lumbar labels caudally by one level, with junction-DSC dropping 29 points on Any-LSTV. No existing CT benchmark surfaces this — not TotalSegmentator's own evaluation, not Li et al. (2025) (the per-class DSC frontier on the matched COLONOG split), and not VERIDAH (vertebra-labeling SOTA on a private CT cohort, operating on pre-localized crops with no pelvis). We contribute: (i) an LSTV-stratified evaluation protocol (per-class DSC, junction-DSC over a 40 mm L5/S1 window, per-class voxel confusion at the junction) translating the level-shift mechanism into quantifiable wrong-level surgical risk; (ii) CTSpinoPelvic1K — 1,153 CT volumes across 802 patients with unified spinopelvic masks, 33 LSTV-positive cases across a 6-way phenotype taxonomy, and radiologist Castellvi typing (I–IV) on all 33; (iii) the first publicly-released merge-based LSTV-handling dual-spinopelvic CT segmenter, exceeding TotalSegmentator on sacrum (TS=0.817 zero-shot) and uniquely delivering both spine and pelvis in a single forward pass with released weights; (iv) an empirical isolation showing VERIDAH's training-side L5/L6 merge eliminates the collision driving TotalSegmentator's level shift but exposes a residual L4/last_lumbar collision on count-style sacralization, motivating VERIDAH's sequence predictor as the necessary disambiguation component. CTSpinoPelvic1K, the protocol, and 5-fold checkpoints are released publicly.

Review Mode: Double-blind (default; anonymized submission)

Dataset Submission: This submission includes a dataset.

Croissant File:  json (/attachment?id=D8yRWDAGQF&name=croissant_file)

Dataset URL: <https://huggingface.co/datasets/anonymous-neurips-ED/CTSpinoPelvic1K>
(<https://huggingface.co/datasets/anonymous-neurips-ED/CTSpinoPelvic1K>)

Dataset Large URL: <https://huggingface.co/datasets/anonymous-neurips-ED/CTSpinoPelvic1K-Sample>
(<https://huggingface.co/datasets/anonymous-neurips-ED/CTSpinoPelvic1K-Sample>)

Code URL: <https://github.com/anonymous-neurips-ED/spinopelvic-seg> (<https://github.com/anonymous-neurips-ED/spinopelvic-seg>)

Code Submission Justification: Submitted

Financial Support:  Gregory John Schwing (/profile?id=~Gregory_John_Schwing1)

Primary Area: AI for health and biotechnology (e.g., digital medicine, computational biology, public health), Computer vision (e.g., object detection, segmentation, image classification, image generation, video)

Contribution Type: Benchmark design and benchmark analysis, e.g., new benchmarks, benchmark redesign, benchmarking methodologies, benchmark saturation or overfitting studies, analyses of benchmark limitations or failure modes.

LLM Usage:  Prefer not to declare (you can then skip the rest)

Other LLM Usage:  None

Author Acknowledgements:  CONTRIBUTION ACKNOWLEDGEMENT: Please confirm that the main paper clearly articulates the evaluative role of the contribution, including what claims it supports, under which assumptions these claims hold, and its limitations. For datasets, the submission should describe how the dataset is intended to be used meaningfully in evaluative practice rather than as an endpoint., CHECKLIST CONFIRMATION: In order to improve the rigor and transparency of research submitted to and published at NeurIPS, authors are required to complete a paper checklist and include it in the paper PDF. See <https://neurips.cc/public/guides/PaperChecklist> (<https://neurips.cc/public/guides/PaperChecklist>) and the provided 2026 sample LaTeX file. Papers not including the checklist in the paper PDF will be desk rejected. Please confirm that a paper checklist is included in the paper PDF., RESPONSIBLE REVIEWING: We acknowledge that every author on the program committee is obliged to conduct responsible ACing / reviewing according to the policy designed by the PC Chairs and relevant members of the organizing committee. Failure to do so may result in desk rejection or other serious consequences., ACADEMIC INTEGRITY: See the NeurIPS Code of Conduct (<https://neurips.cc/public/CodeOfConduct> (<https://neurips.cc/public/CodeOfConduct>)) and the NeurIPS policy of Academic Integrity in Research Paper Submissions and Reviews (<https://neurips.cc/Conferences/2026/ReviewerIntegrity> (<https://neurips.cc/Conferences/2026/ReviewerIntegrity>)). I acknowledge that I have read and commit to adhering to the NeurIPS Code of Conduct and the NeurIPS Academic Integrity Policy., DECLARATION: I confirm that the information provided in this form is accurate.

Reviewer Nomination:  Loren Schwiebert (/profile?id=~Loren_Schwiebert1)

Submission Number: 3474

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9 / 9 replies shown

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Withdrawal

Paper Decision

Decision by Program Chairs  24 Sept 2026, 19:25 (modified: 24 Sept 2026, 19:41)

 Program Chairs, Authors  Revisions (/revisions?id=hLCLqbRkKb)

Decision: Accept (poster)

Metareview:

(a) Summary. The paper audits TotalSegmentator on Lumbosacral Transitional Vertebrae: lacking an L6 class, it shifts lumbar labels one level caudally on lumbarization cases, resembling a wrong-level plan. It releases CTSpinoPelvic1K (1,153 CTs) with an LSTV taxonomy + Castellvi typing, a stratified protocol, and a merge baseline.

(b) Strengths. A clinically important, well-motivated audit of the most deployed system; a stratified protocol with junction metrics that aggregate scores hide; honest scoping and public release.

(c) Weaknesses. The merge advantage is inferred only against a system that structurally lacks L6, so it isn't isolated from added data + an L6 class (aoYn, jbUM). Only 33 LSTV cases (test 7; conspicuous-biased, 1 Type I); labels from two residents with no inter-rater agreement or senior adjudication; the wrong-level claim is unvalidated on workflow. Fbco (conf 2) questions the comparison — partly a misread, but answer it.

(d) To address in rebuttal. Add a same-data unmerged baseline (ideally an L6-retrained model) to isolate the merge; report per-subgroup uncertainty and restrict claims to the conspicuous regime; report inter-rater agreement (kappa) + senior adjudication; soften the wrong-level claim to a plausible risk mechanism unless reader/planning evidence is added.

(e) I read the response. The paper has two layers. A segmenter with no L6 class shifts lumbar labels one level caudally, and rerunning it on released volumes reproduces this without any contested read; the unmerged same-data control I required still calls 84.7 to 99.4 percent of ground-truth L6 an L5, so the merge is the cause. The characterisation layer is weaker, as you disclose: subtypes from filename heuristics rather than image reads, Castellvi typing by two-resident consensus with no independent scoring, 7 LSTV cases in the held-out split. The reproducible layer carries the paper. At camera-ready, state the label provenance and its limits, scope LSTV-stratified claims to the test-split n, and report the subtype counts. New data, more baselines and a reader study are not held against you.

(f) I posted no comments to the reviewers. Only aoYn posted after the rebuttal, accepting the unmerged control while holding its rating on the annotation layer. Fbco moved from 3 to 4 and jbUM held at 5, neither posting.

Final Justification:

Final ratings are 3, 4 and 5, ranking 4/10 in my pool, with the highest score from the most confident reviewer.

The audit holds. A segmenter with no L6 class shifts lumbar labels one level caudally, and a reader can reproduce that on released volumes. The unmerged same-data control isolates the merge. The LSTV characterisation layer on top is what the paper does not establish.

Label provenance. The six-way subtypes are heuristic, the Castellvi layer is a two-trainee consensus with no independent read, and the test split holds 7 LSTV cases. Disclosure and scoping are camera-ready conditions.

Review of Submission 3474

Official Review by Reviewer Fbco 📅 21 Jul 2026, 05:30 (modified: 24 Sept 2026, 19:51)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Fbco

📄 Revisions (/revisions?id=WJSYyPOXgA)

Summary:

This paper identifies that TotalSegmentator systematically shifts lumbar labels caudally by one level on lumbarization cases due to its missing L6 class, producing outputs indistinguishable from wrong-level surgical plans without any error signal. To address this evaluation gap, the authors propose an LSTV-stratified protocol with four clinically interpretable metrics, release CTSPINOPELVIC1K, a benchmark of 1,153 CTs with six-way LSTV taxonomy and Castellvi annotations, and provide a merge-based baseline that empirically isolates the necessity of a sequence predictor beyond label merging.

Contribution Type Check: Yes

Strengths:

- The paper identifies a clinically critical, silent failure mode in TotalSegmentator that maps directly to wrong-level surgical risk, and introduces the first LSTV-stratified evaluation protocol that translates this level-shift mechanism into quantifiable surgical metrics.

- The paper contributes CTSPINOPELVIC1K, a publicly released benchmark of 1,153 CT volumes with unified spinopelvic masks, a six-way LSTV taxonomy, and Castellvi typing for all LSTV cases, while resolving a previously undocumented coordinate mismatch and taxonomic bug.
- The paper provides the first empirical ablation isolating the effect of merge, showing that the merge eliminates TotalSegmentator's collision but leaves a residual collision that requires a sequence predictor, and its baseline surpasses TotalSegmentator with all code and checkpoints publicly released.

Weaknesses:

- Although the paper constructs a new dataset CTSPINOPELVIC1K, it lacks a detailed comparisons over other datasets to highlight its contributions. It remains unclear how the proposed dataset or evaluation protocol will inspire future studies. Moreover, it is better to add an illustration figure to better demonstrate the overall construction process.
- The paper lacks sufficient background or preliminary details to make it easier to understand. Currently, it excludes many concepts essential to ease the understanding. Moreover, some illustrations are not well-organized, e.g., both the introduction and the related work sections should pay more attention to the content relevant to the topic of the paper rather than list some generic descriptions.
- The experimental setting of TotalSegmentator seems to be problematic. If the label vocabulary of TotalSegmentator does not contain L6 class, it is reasonable that it does not perform well. Therefore, it is unclear the reason to such methods, and since there lacks sufficient discussions of related work, it is unclear whether there exist some other methods more appropriate for the experiment comparison and analysis (or more methods should be included to support the results).

Ethical Concerns: No or very minor ethics concerns only

Reproducibility: Yes

Dataset Assessment: Yes

Limitations:

The limitations have been listed.

Questions:

- How will the proposed dataset or evaluation protocol inspire future studies beyond serving as a validation set for the proposed baseline?
- Could the authors provide a systematic comparison with other datasets and add an illustrative figure to better demonstrate the construction pipeline?
- Could the authors expand the background and preliminaries (e.g., LSTV subtypes, Castellvi classification, VERIDAH architecture) and reorganize the introduction/related work to focus more on the paper's specific topic?
- Given that TotalSegmentator lacks an L6 class, its poor performance is structurally expected; why is this comparison meaningful, and should other existing methods (e.g., SPINEPS or a retrained variant with L6) be included to strengthen the analysis?

Rating: 4: Borderline accept: Technically solid paper where reasons to accept outweigh reasons to reject, e.g., limited evaluation. Please use sparingly.

Confidence: 2: You are willing to defend your assessment, but it is quite likely that you did not understand the central parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.

Paper Formatting Concerns:

no

Final Justification:

The authors have provided rebuttals to address some of the concerns.

LLM Policy: Yes

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

**Rebuttal by Authors**

Rebuttal

by Authors (👁 Miraziz Ismoilov (/profile?id=~Miraziz_Ismoilov1), Patrick Schwing (/profile?id=~Patrick_Schwing1), Nizar Alnabahneh (/profile?id=~Nizar_Alnabahneh1), Loren Schwiebert (/profile?id=~Loren_Schwiebert1), +1 more (/group/edit?id=NeurIPS.cc/2026/Evaluations_and_Datasets_Track/Submission3474/Authors))

📅 27 Jul 2026, 17:00 (modified: 28 Jul 2026, 08:31)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

📄 Revisions (/revisions?id=7HTrf7dXc6)

Rebuttal:

Thank you. We clarify the comparison and add the requested methods and figures.

"TS lacks L6, so the comparison isn't fair": TS is the most-used CT segmenter and 3D Slicer's default back-end, so its output is what clinicians get today, and the failure is silent (a plausible, one-level-off labeling with no error flag). More importantly (General Response AC-1), this is not a TS-specific quirk. Our own model, trained with an L6 class on the same data, makes the identical L6 to L5 error. So the comparison reveals a general limitation of dense segmentation on LSTV, not a handicap unique to TS.

Other methods (SPINEPS, retrained-with-L6): The L6-retrained unmerged model described above is exactly the "retrained variant with L6" you suggested. We will additionally report SPINEPS evaluated under the same protocol in the revised paper.

Comparison to other datasets and future use: We add the table below. No existing public CT dataset provides unified spine and pelvis, LSTV stratification, and Castellvi typing together. On future use, our L6 result directly motivates counting-aware and two-stage "segment then number" architectures. Because the L5 vs L6 cue is a count that exceeds a convolutional patch and is data-starved, adding segmentation capacity is the wrong lever, and this benchmark is the natural test set for that work.

Dataset	Spine	Pelvis	Unified cohort	LSTV-stratified	6-way phenotype	Castellvi typing	Public
VerSe	✓						✓
CTSpine1K	✓						✓
TotalSegmentator	✓ (no L6)	✓	partial				✓
CTPelvic1K		✓					✓
VERIDAH cohort	✓						
CTSpinoPelvic1K (ours)	✓	✓	✓	✓	✓	✓ (all 33)	✓

Background and figure: We add a Background section (LSTV subtypes; Castellvi I to IV; the VERIDAH architecture) and a dataset-construction figure, and we refocus the introduction and related work on transitional-vertebra labeling.

Vwertebrae

Official Review

by Reviewer jbUM

📅 25 Jun 2026, 23:12 (modified: 24 Sept 2026, 19:51)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer jbUM

📄 Revisions (/revisions?id=lr1ImUiMvp)

Summary:

The paper studies a clinically important failure mode of TotalSegmentator on lumbosacral transitional vertebrae (LSTV). The authors show that because TotalSegmentator lacks an L6 class, lumbarization cases can produce a systematic one-level caudal label shift that resembles the mechanism of wrong-level spine surgery. They introduce

CTSPINOPELVIC1K, a spinopelvic CT benchmark with LSTV stratification, Castellvi typing, and metrics focused on the L5/S1 junction. The main contribution is showing that aggregate spine segmentation scores can hide safety-critical subgroup failures, and that LSTV-specific evaluation is necessary for surgical-planning applications.

Contribution Type Check: Yes

Strengths:

- Very clinically important failure mode
- Clear wrong-level surgery motivation
- Strong safety-critical framing
- Direct audit of deployed TotalSegmentator
- LSTV-stratified evaluation protocol
- Useful junction-specific DSC metric
- Strong case-study visualization
- New spinopelvic CT benchmark
- Includes Castellvi annotation layer
- Evaluates phenotype-specific failures
- Good analysis of label-shift mechanism
- Public data and code release
- Released 5-fold checkpoints promised
- Strong argument against aggregate-only reporting
- Practical relevance for surgical planning

Weaknesses:

- Only 33 LSTV-positive patients
- Held-out LSTV test set is very small
- No semi-sacralization case in test split
- Castellvi labels lack inter-rater agreement
- Normal cohort not screened for Type I
- Dataset biased toward conspicuous LSTV
- Strong dependence on inherited heuristics
- Requires H200-class hardware
- No same-compute unmerged baseline
- TotalSegmentator evaluated only zero-shot
- Comparison to VERIDAH is indirect
- Merge-based segmentation remains insufficient
- Upper-lumbar label shift persists
- Clinical workflow validation is not shown
- Surgical-risk claim may need careful wording

Ethical Concerns: No or very minor ethics concerns only

Reproducibility: Yes

Reproducibility Comments:

data is online

Dataset Assessment: Yes

Dataset Comments:

yes

Limitations:

yes

Questions:

Could the authors clarify the intended strength of the wrong-level surgery claim? The TotalSegmentator output clearly creates a one-level label-shift pattern on lumbarization, but the paper does not evaluate actual surgical planning workflows or clinician behavior. I would suggest framing this as a plausible automated wrong-level-risk mechanism unless supported by reader studies, planning simulations, or downstream workflow evidence.

Rating: 5: Accept: Technically solid paper, with high impact on at least one sub-area of AI or moderate-to-high impact on more than one area of AI, with good-to-excellent evaluation, resources, reproducibility, and no unaddressed ethical considerations.

Confidence: 5: You are absolutely certain about your assessment. You are very familiar with the related work and checked the math/other details carefully.

Paper Formatting Concerns:

n/a

Final Justification:

thank you for the clarifications. score holds

LLM Policy: Yes

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes



Rebuttal by Authors

Rebuttal

by Authors (Miraziz Ismoilov (/profile?id=~Miraziz_Ismoilov1), Patrick Schwing (/profile?id=~Patrick_Schwing1), Nizar Alnabahneh (/profile?id=~Nizar_Alnabahneh1), Loren Schwiebert (/profile?id=~Loren_Schwiebert1), +1 more (/group/edit?id=NeurIPS.cc/2026/Evaluations_and_Datasets_Track/Submission3474/Authors))

27 Jul 2026, 17:00 (modified: 28 Jul 2026, 08:31)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Revisions (/revisions?id=n5QAtudBdQ)

Rebuttal:

Thank you for the strong support and the precise framing.

Wrong-level wording: We adopt your exact framing, a plausible automated wrong-level-risk mechanism, and add a "what we do and do not show" sentence (General Response AC-4).

"No same-compute unmerged baseline": Now directly addressed (General Response AC-1). Same compute, same folds, with an L6 class and oversampling, and it still labels 84.7% to 99.4% of GT-L6 as L5. We scope this as a limitation of dense segmentation at the current LSTV data size (the L5/L6 count is non-local and data-starved), not an impossibility, and we point to counting-aware and two-stage designs as future work.

A clinically important audit showing that TotalSegmentator systematically mislabels lumbosacral transitional vertebrae, paired with a new public spinopelvic CT benchmark (CTSpinoPelvic1K) and a baseline segmenter, although the central LSTV findings rest on a small and conspicuously biased set of 33 cases.

Official Review by Reviewer aoYn 23 Jun 2026, 12:34 (modified: 24 Sept 2026, 19:51)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer aoYn

Revisions (/revisions?id=vo0wllfMaq)

Summary:

This paper audits TotalSegmentator, the most widely deployed CT segmentation system, and shows that it systematically fails on the Lumbosacral Transitional Vertebra (LSTV, 5 to 35 percent prevalence): it lacks an L6 class and shifts lumbar labels one level caudally, producing an output that is locally well formed but anatomically one level off. To study and address this, the authors release CTSpinoPelvic1K, a benchmark of 1,153 CT records from 802 patients that fuses COLONOG with CTSpine1K vertebra and CTPelvic1K pelvic annotations and adds a six-way LSTV taxonomy and Castellvi typing on 33 LSTV cases, an LSTV-stratified evaluation protocol, and a merge-based dual-spinopelvic baseline that exceeds TotalSegmentator on the sacrum. However, while the clinical motivation and the

openly released benchmark are valuable, there are concerns regarding the small and conspicuously biased LSTV sample of 33 cases, the verification of the new labels by only two trainees without senior adjudication, and the absence of a same-data ablation isolating the merge, which the authors should address.

Contribution Type Check: Yes

Strengths:

The problem is clinically important and well motivated: wrong-level spine surgery is a persistent never-event, LSTV is its largest anatomic driver, and auditing the most widely deployed CT segmentation system on this failure is high-impact. The failure-mode analysis is rigorous and mechanistic, explaining the caudal level-shift, showing that L6 is structurally absent from the label vocabulary, and translating the mechanism into a concrete wrong-level surgical risk with a clean case study. The benchmark fills a real gap, fusing spine and pelvic annotations on a shared cohort with LSTV stratification, a six-way phenotype taxonomy, and a Castellvi typing layer that existing public CT benchmarks do not provide together. The release is reproducibility-minded, with Croissant metadata, Responsible AI fields, a sample subset, a permissive license, and planned checkpoints. Finally, the scoping is honest and self-critical, isolating the contribution of the merge from a sequence predictor and acknowledging that the merge is necessary but not sufficient.

Weaknesses:

1. The entire LSTV evaluation rests on only 33 LSTV-positive cases, several subtypes have just a handful (e.g., count-style sacralization $n=9$, semi-sacralization $n=2$), the held-out test split has only 7 of 33, and the paper reports a selection bias toward conspicuous presentations (only 1 of 33 is the subtler Castellvi Type I), so the headline numbers characterize the conspicuous regime rather than the full LSTV population. Therefore, the authors should report uncertainty for the small subgroups, expand the rare phenotypes (especially Type I), and restrict the quantitative claims to the conspicuous regime that the 33 cases actually cover.
2. The new LSTV and Castellvi annotation layers that give the benchmark its value are verified only by two resident radiologists by consensus, with no inter-rater scoring and no senior adjudication (which the paper acknowledges), so confidence in the ground truth and in every downstream LSTV-stratified result is limited. Therefore, the authors should report an inter-rater agreement metric (e.g., Cohen's kappa) and add senior-radiologist adjudication on at least a representative subset of the 33 cases.
3. The advantage attributed to the L5/L6 merge is inferred from a zero-shot comparison to TotalSegmentator, which structurally lacks an L6 class, rather than from a same-data unmerged baseline, so the benefit of the merge cannot be separated from that of the new data and the added L6 class. Therefore, the authors should add a same-data control, namely an unmerged baseline trained on CTSpinoPelvic1K and ideally a TotalSegmentator-style model retrained with an L6 class, to isolate the merge's contribution.
4. The residual L4/last_lumbar collision that motivates the fourth contribution shows high cross-fold variability (Table 6 reports an L4 Dice of 0.187 with a standard deviation of 0.337, driven by one outlier fold) and pools correlated voxels across only a handful of sacralization cases, so its statistical basis is weak. Therefore, the authors should report case-level rather than voxel-level uncertainty and confirm the finding on substantially more sacralization cases.

Ethical Concerns: No or very minor ethics concerns only

Reproducibility: Partly

Dataset Assessment: Yes

Limitations:

Limitations are discussed thoroughly and candidly.

Questions:

1. The authors should report uncertainty for the small LSTV subgroups, expand or augment the rare phenotypes (especially Castellvi Type I), and restrict the claims to the conspicuous regime that the 33 cases actually cover.
2. The authors should report inter-rater agreement (for example Cohen's kappa) for the new LSTV and Castellvi labels and add senior-radiologist adjudication on at least a subset.
3. The authors should add a same-data control, namely an unmerged baseline trained on CTSpinoPelvic1K and ideally a TotalSegmentator-style model retrained with an L6 class, so that the benefit of the merge can be separated from the benefit of the new data and the added L6 class.
4. The authors should report case-level rather than voxel-level uncertainty for the residual L4/last_lumbar collision and confirm the finding on a larger number of sacralization cases.

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Paper Formatting Concerns:

No violations observed.

Final Justification:

I thank the authors for the rebuttal. The unmerged control is the right control, and Question 3 is answered. Training a same-data, same-fold, same-architecture model with an explicit L6 class and L6 oversampling, and still finding that 84.7 to 99.4 percent of ground-truth L6 is labelled L5, does isolate the merge from the extra data and from the added class.

I am nevertheless maintaining my rating of 3.

Question 3 is the only point answered with a new experiment, and the remaining three answers change what is claimed rather than what is shown. Question 2 is answered by a plan. The senior adjudication is described as arranged and the agreement statistic as something the revised paper will report, so there is no kappa value and no adjudication outcome available to me. I weigh this most heavily, because most of the 1,153 records are inherited from COLONOG, CTSpine1K, and CTPelvic1K, so what distinguishes this benchmark from its upstream sources is concentrated in the 33 annotated cases. The authors further state that the adjudication may produce corrections in the released dataset, which means those labels are expected to change. Every LSTV-stratified result therefore remains provisional, including the new unmerged control, whose ground-truth L6 comes from the same annotation layer. Question 1 is answered by reporting n and confidence intervals, whose values are not given here, and by placing the single Castellvi Type I case out of scope, which narrows the claims rather than enlarging the evidence. The 33-case sample is unchanged. Question 4 likewise re-presents the existing L4 number with caveats rather than confirming it on more sacralization cases. I would add that the held-out test split contains no semi-sacralization case, so per-subgroup reporting will show that subtype as unevaluated rather than measured, and the restriction of the claims should say so explicitly.

LLM Policy: Yes

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes



Rebuttal by Authors

Rebuttal

by Authors (Miraziz Ismoilov (/profile?id=~Miraziz_Ismoilov1), Patrick Schwing (/profile?id=~Patrick_Schwing1), Nizar Alnabahneh (/profile?id=~Nizar_Alnabahneh1), Loren Schwiebert (/profile?id=~Loren_Schwiebert1), +1 more (/group/edit?id=NeurIPS.cc/2026/Evaluations_and_Datasets_Track/Submission3474/Authors))

27 Jul 2026, 16:51 (modified: 28 Jul 2026, 08:31)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Revisions (/revisions?id=WXsOoa91LV)

Rebuttal:

Thank you. We addressed all four points.

W3 (isolate the merge): Done, see General Response AC-1. A same-data, same-fold, same-architecture unmerged model with an L6 class and L6-oversampling still labels 84.7% to 99.4% of GT-L6 as L5, isolating the merge from the extra data and the added class. We scope this as "not learned by a standard dense segmenter at this data size" (recoverable via more data or a counting-aware or two-stage design, future work), not an impossibility. The reason is architectural: the L5 vs L6 signal is a count that spans more than one convolutional patch and appears in very few examples.

W1 (few, mostly-obvious cases): We now show n next to every subgroup number and add 95% confidence intervals on the Dice scores, and we limit our quantitative claims to the obvious (Castellvi II to IV) cases; the single Type I case is out of scope for quantitative claims.

W2 (only residents): We have arranged independent senior radiologist adjudication of all 33 LSTV cases. The revised paper will report inter-rater agreement (Cohen's kappa) with the resident labels and the resulting corrections in the released dataset.

W4 (weak L4 number): We report it per case rather than pooled, note it is driven by one fold and only 9 cases, and present it as a directional motivation for the sequence predictor, not a precise value.



➔ *Replying to Rebuttal by Authors*

Official Comment by Reviewer aoYn

Official Comment by Reviewer aoYn 📅 03 Aug 2026, 09:45

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

I thank the authors for the rebuttal. The unmerged control is the right control, and Question 3 is answered. Training a same-data, same-fold, same-architecture model with an explicit L6 class and L6 oversampling, and still finding that 84.7 to 99.4 percent of ground-truth L6 is labelled L5, does isolate the merge from the extra data and from the added class.

I am nevertheless maintaining my rating of 3.

Question 3 is the only point answered with a new experiment, and the remaining three answers change what is claimed rather than what is shown. Question 2 is answered by a plan. The senior adjudication is described as arranged and the agreement statistic as something the revised paper will report, so there is no kappa value and no adjudication outcome available to me. I weigh this most heavily, because most of the 1,153 records are inherited from COLONOG, CTSpine1K, and CTPelvic1K, so what distinguishes this benchmark from its upstream sources is concentrated in the 33 annotated cases. The authors further state that the adjudication may produce corrections in the released dataset, which means those labels are expected to change. Every LSTV-stratified result therefore remains provisional, including the new unmerged control, whose ground-truth L6 comes from the same annotation layer. Question 1 is answered by reporting n and confidence intervals, whose values are not given here, and by placing the single Castellvi Type I case out of scope, which narrows the claims rather than enlarging the evidence. The 33-case sample is unchanged. Question 4 likewise re-presents the existing L4 number with caveats rather than confirming it on more sacralization cases. I would add that the held-out test split contains no semi-sacralization case, so per-subgroup reporting will show that subtype as unevaluated rather than measured, and the restriction of the claims should say so explicitly.

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